

Technical Document #293: Cybersecurity - Comprehensive Overview

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Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Social engineering manipulates people into revealing confidential information.
- Incident response plans define how organizations respond to security incidents.
- Penetration testing simulates cyber attacks to identify vulnerabilities.